

Apache Camel EIP Cheat Sheet

The Camel 4 DSL for every Enterprise Integration Pattern in this course — routing, transformation, reliability, endpoints, testing, and the run commands — on one page.

Print this page (Cmd/Ctrl+P) for a physical card. New here? Start with the [guide](#), then work the [curriculum](#) lesson by lesson.

Group 01

Route basics

RouteBuilder.java

```
from("direct:orders")
    .routeId("order-intake")
    .log("received ${body}")
    .to("bean:validate")
    .to("seda:accepted");

// a timer-driven source
from("timer:tick?period=5000")
    .routeId("feed")
    .to("direct:orders");
```

Covered in: [03](#) · [04](#)

Group 02

Simple language

Simple expressions

```
${body}           // the message body
${header.country}  // one header (also headers.country)
${body.total}      // bean property → getTotal()
${exchangeProperty.count}
```

```
// predicates (use inside simple("..."))
${header.country} == 'US'
${body.total} > 100
${header.sku} regex '[A-Z]{3}-\d+'
${body} contains 'urgent'
```

Covered in: [05](#) · [09](#) · [10](#)

Group 03

Routing EIPs

Routing DSL

```
.choice()
  .when(simple("${header.country} == 'US'")).to("direct:us")
  .when(simple("${header.country} == 'UK'")).to("direct:uk")
  .otherwise().to("direct:intl")
.end()

.filter(simple("${body.total} > 0")).to("direct:billable")

.split(body()).to("direct:line")      // one → many
.split(jsonPath("$.items"))
.split(body().tokenize(", "))        // split a CSV body

.aggregate(header("orderId"),
  new GroupedBodyAggregationStrategy(),
  .completionSize(10).completionTimeout(2000)

.recipientList(header("recipients")) // comma-separated URIs
.multicast().parallelProcessing()
  .to("direct:a", "direct:b").end()
.wireTap("direct:audit")              // fire-and-forget copy
.dynamicRouter(method(Router.class, "next"))
.routingSlip(header("itinerary"))
```

Covered in: [08](#) · [09](#) · [10](#) · [11](#) · [12](#) · [13](#) · [14](#)

Group 04

Transformation

Transform & marshal

```
.transform(simple("Hello ${body.name}"))
.setBody(simple("${body.customer}"))
.setHeader("region", simple("${body.country}"))

// JSON to/from POJO (Jackson)
.unmarshal().json(JsonLibrary.Jackson, Order.class)
.marshal().json(JsonLibrary.Jackson)

// call a bean
.bean(OrderService.class, "enrich")
.to("bean:orderService?method=score")

// content enricher – merge in extra data
.enrich("direct:lookupCustomer", new MergeStrategy())
.pollEnrich("file:rates?fileName=fx.json")
```

Covered in: [15](#) · [16](#)

Group 05

Reliability

Error handling

```
// global dead-letter + redelivery / backoff
errorHandler(deadLetterChannel("seda:dead")
    .maximumRedeliveries(3)
    .redeliveryDelay(1000)
    .backOffMultiplier(2)
    .useOriginalMessage());

// per-exception policy
onException(IOException.class)
    .maximumRedeliveries(2)
    .handled(true)
    .to("direct:failed");

// process each id at most once
from("direct:in")
    .idempotentConsumer(header("orderId"),
        MemoryIdempotentRepository
```

```
        .memoryIdempotentRepository(500))  
        .to("direct:process");
```

Covered in: [07](#) · [19](#) · [20](#)

Group 06

Endpoints

Endpoint URIs

```
direct:name      // sync · same thread · same JVM  
seda:name        // async in-memory queue (staged)  
vm:name          // async · across contexts in one JVM  
  
jms:queue:orders // point-to-point (one consumer)  
jms:topic:events // publish-subscribe (all subscribers)  
kafka:orders?brokers=localhost:9092  
  
timer:tick?period=5000  
file:inbox?move=.done
```

Covered in: [04](#) · [18](#)

Group 07

Testing

RouteTest.java

```
@CamelSpringBootTest  
@SpringBootTest  
class RouteTest {  
    @Autowired CamelContext context;  
    @Autowired ProducerTemplate template;  
    @EndpointInject("mock:result") MockEndpoint mock;  
  
    @Test void routesToResult() throws Exception {  
        mock.expectedMessageCount(1);  
        mock.expectedBodiesReceived("OK");  
        template.sendBody("direct:in", "ping");  
        mock.assertIsSatisfied();  
    }  
}
```

```
}  
}  
  
// swap a real endpoint for a mock at test time  
AdviceWith.adviceWith(context, "order-intake",  
    a -> a.weaveById("send").replace().to("mock:result"));  
  
// assert message flow without mocks  
NotifyBuilder done = new NotifyBuilder(context)  
    .whenDone(1).create();
```

Covered in: [06](#)

Group 08

Commands

terminal

```
./mvnw test                # unit tests only  
./mvnw verify              # full build + integration  
./mvnw -pl patterns/10-content-based-router test  
./mvnw -pl patterns/10-content-based-router spring-boot:run  
./mvnw -P upstream verify  # plain Apache Camel (Central)  
make up                   # start the broker (lesson 18)
```

Covered in: [Setup](#) · [18](#)

TIP These snippets are trimmed to the essential DSL. Every group maps to a runnable lesson module — open the linked lesson to see the full route, its test, and the reasoning. See the [guide](#) for how to read a lesson, or jump into the [curriculum](#).